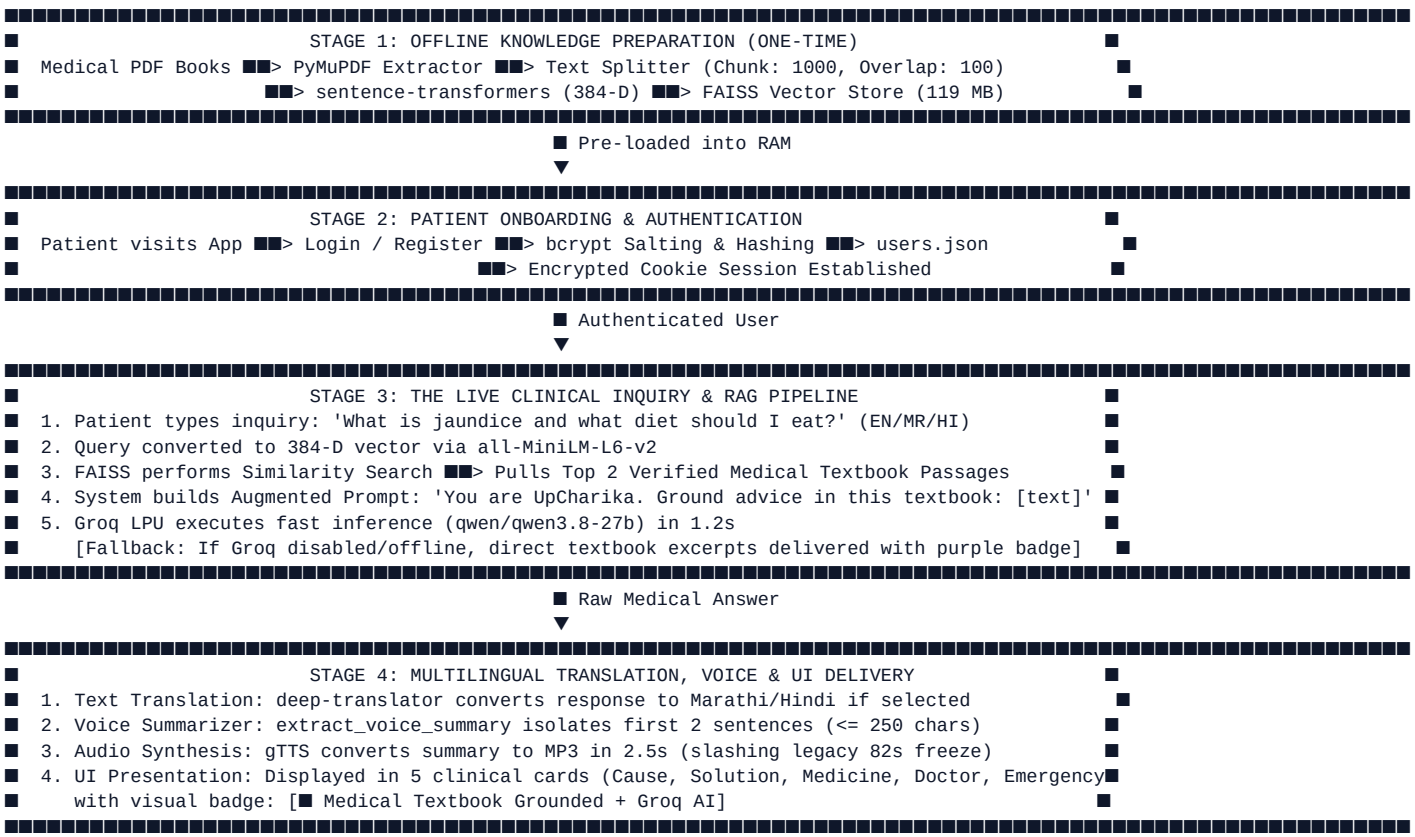
UpCharika: Complete Project Flow & Lifecycle

Executive Summary: How Every Part of the System Works from Start to Finish



Stage 1: Offline Knowledge Preparation (The Foundation)

Before the chatbot ever starts, the medical knowledge must be transformed into a searchable mathematical matrix. This is handled by create_memory_for_llm.py:

• 1. Document Extraction: Uses fitz (PyMuPDF) to read clinical textbook PDFs page-by-page.

• 2. Intelligent Chunking: Splits long text into 1,000-character segments using RecursiveCharacterTextSplitter. It maintains a 100-character overlap between adjacent segments so sentences are never cut in half.

• 3. Vector Embeddings: Passes each chunk through sentence-transformers/all-MiniLM-L6-v2, converting textual meaning into 384 decimal numbers.

• 4. FAISS Indexing: Saves all vectors into vectorstore/db_faiss/index.faiss (119 MB), enabling sub-20ms nearest-neighbor semantic search.

Stage 2: User Access & Clinical Security (The Gatekeeper)

Healthcare applications require secure, tamper-proof user sessions:

• 1. Salted Bcrypt Hashing: When a user signs up, their password is encrypted using bcrypt with 12 algorithmic salt rounds. Passwords cannot be reverse-engineered even if users.json is accessed.

• 2. Persistent Sessions: Uses streamlit-cookies-manager with AES encryption to store login credentials in browser cookies, keeping the patient logged in even across page reloads.

• 3. Seamless Sign Out: Clicking 'Sign Out' flushes session state and clears encrypted cookies instantly.

Stage 3: The Live Clinical Inquiry & True RAG Pipeline (The Core Brain)

When a patient submits a question (e.g. 'What is jaundice and what diet should I eat?'), medibot.py processes it through True RAG:

• 1. Vectorization: The patient's query is converted into a 384-dimensional vector using the exact same MiniLM embedding model.

• 2. FAISS Similarity Search: FAISS calculates Cosine Similarity across the 119 MB index and extracts the Top 2 verified textbook passages.

• 3. Prompt Augmentation: The retrieved textbook paragraphs are injected into a strict system prompt instructing the LLM: 'Base your clinical guidance exclusively on this textbook evidence.'

• 4. Ultra-Fast Groq Inference: Groq LPUs run qwen/qwen3.8-27b (with fallback to openai/gpt-oss-20b) with temperature=0.3, delivering answers in 1.2s.

• 5. Fault Tolerance & Test Mode: If Groq is disabled or offline, the system delivers direct textbook excerpts with the purple badge: 100% Direct Textbook (Groq OFF).

Stage 4: Multilingual Translation, Voice & Clinical Delivery (The Output)

The generated clinical answer is converted into accessible patient triage:

• 1. Regional Translation: If the patient selected Marathi or Hindi, the response is translated using native Devanagari templates or deep-translator (Google Engine).

• 2. Voice Synthesis Latency Optimization: extract_voice_summary extracts the first 2 actionable sentences (<= 250 chars). gTTS synthesizes audio in 2.5 seconds (eliminating the legacy 82s document freeze).

• 3. Standardized 5-Section Cards: The answer is displayed in clean, color-coded clinical cards: Cause (Orange), Solution / Diet (Green), Medicine (Teal), Doctor (Purple), and Emergency (Red).

• 4. Visual Grounding Badges: Every message displays a badge showing its exact origin: Medical Textbook Grounded + Groq AI, 100% Direct Textbook (Groq OFF), or Offline Clinical KB.

Summary Matrix of Project Technologies Across All 4 Stages

Lifecycle Stage	Components & Libraries	Exact Function in UpCharika
Stage 1: Prep	PyMuPDF (fitz), LangChain TextSplitter, sentence-transformers, FAISS	Extracts PDFs, splits into 1000-char chunks (overlap 100), builds 119 MB vector index.
Stage 2: Auth	bcrypt, EncryptedCookieManager, users.json	Encrypts passwords with 12 salt rounds; maintains persistent client cookies.
Stage 3: RAG	FAISS search, Groq LPUs, qwen/qwen3.8-27b	Finds top 2 textbook paragraphs, augments prompt, generates factual clinical triage in 1.2s.
Stage 4: Output	deep-translator, extract_voice_summary, gTTS, Streamlit, style.css	Translates to Marathi/Hindi, generates 2.5s audio, renders 5-tier clinical triage cards.